

SC2001

ALGORITHM DESIGN AND ANALYSIS

Project 2: The Dijkstra's Algorithm

In the Dijkstra's algorithm, the choice of the input graph representation and the priority queue implementation will affect its time complexity.

- (a) Suppose the input graph $G = (V, E)$ is stored in an *adjacency matrix* and we use an *array* for the priority queue. Implement the Dijkstra's algorithm using this setting and analyze its time complexity with respect to $|V|$ and $|E|$ both theoretically and empirically.
- (b) Suppose the input graph $G = (V, E)$ is stored in an *array of adjacency lists* and we use a *minimizing heap* for the priority queue. Implement the Dijkstra's algorithm using this setting and analyze its time complexity with respect to $|V|$ and $|E|$ both theoretically and empirically.
- (c) Compare the two implementations in (a) and (b). Discuss which implementation is better and in what circumstances.